How the Programme Fits Together: One Family of Polytopes, Many Models, and One Example Through Every Layer

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Abstract

The programme consists of eight papers: a posterior tower, a categorical account of conditionals and couplings, a companion on coupling polytopes, a paper on three layers of data, one on precise attribution, one on coupling algebras, one on bands of probabilities, and a note on bridges to entropy and branes. This note explains how they fit together. The common object is not a single category or a single geometric realization. It is the family of coupling polytopes, the fibres of the marginal map over the space of margins, together with its stratification by walls and by the boundary of the simplex. Each paper puts one kind of structure on this family. Several mathematical languages model it, and each sees something the others miss: linear programming, sheaves and descent, simplicial sets, cumulants and A_∞-structures, information geometry, toric and birational geometry, Floer theory, and stability conditions. Floer theory is one of these models, not the privileged one. We test a proposed unification by deformation theory and find that it needs one correction. The natural algebraic and complex-geometric objects are rigid: the gluing algebra is separable, the fibres are affine, and every 2×3 toric surface has $H^1(T_X) = H^2(T_X) = 0$. What moves with the margins is the symplectic class, which is the product coupling. Its deformation space is the space of Kähler classes, cut into chambers by the secondary fan, and the bending, tearing and breaking of the bands paper are movement inside a chamber, crossing a wall, and reaching the boundary of the effective cone. One 2×3 example is followed through every layer.

Contents

1 Why this note **2**

2 The common object **2**

3 What each paper does **4**

4 Many models of one family **4**

*All papers of the programme: https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/papers. Code and checks: https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/architecture. `check_architecture.py` recomputes every number in sections 5 and 6. The other results quoted here are proved or computed, with their own checks, in the papers cited.

5 Deformation theory: what it unifies and what it does not **5**
5.1 The natural candidates are rigid 5
5.2 What does deform: the symplectic class 6
5.3 The Floer side has its own deformation space 7

6 One example through every layer **7**

7 Status **9**

8 Computational checks **10**

1 Why this note

A reader who meets the programme through one paper sees one structure: a tower of posteriors, a 2-cell, a polytope, a band, a brane. The papers cite each other, but the reason they belong together is easy to miss. Two framings have been proposed to make it visible. One says that Floer theory is only one of many possible realizations of the programme's structure. The other says that all the papers study one deformation-theoretic object, a cotangent complex of the couplings. We keep the first and correct the second.

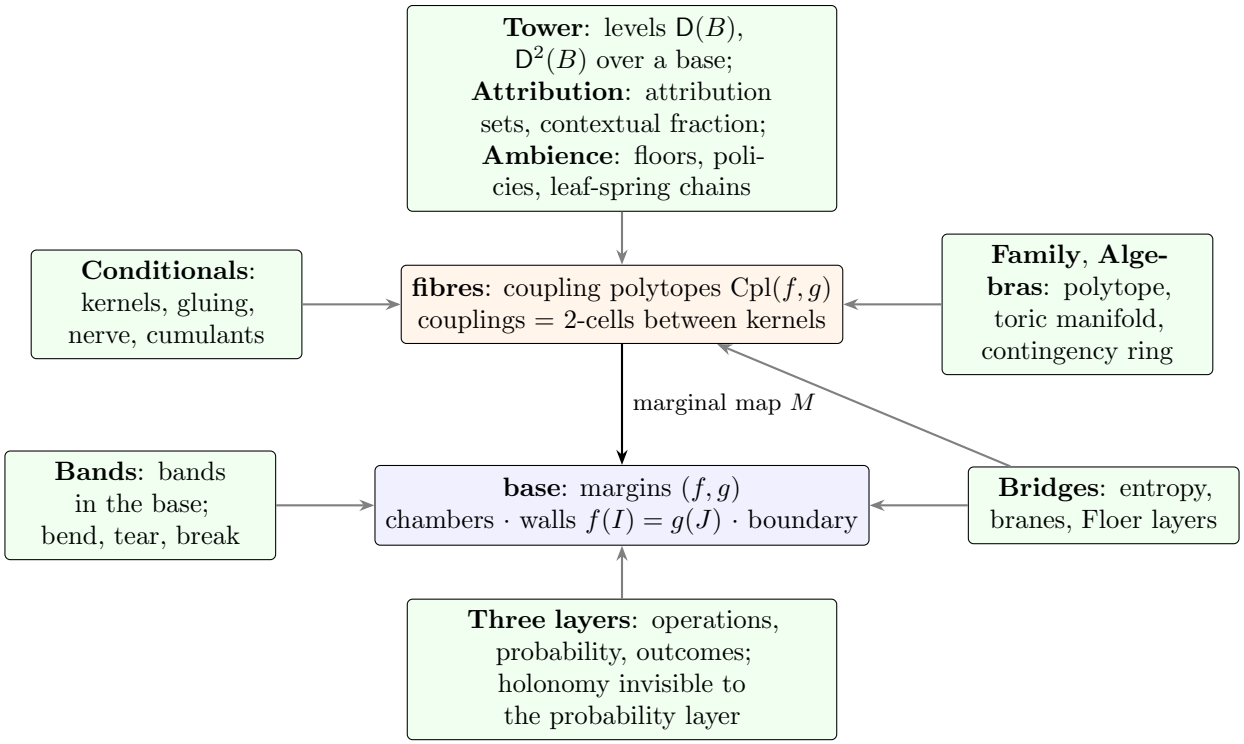
The plan is as follows. Section 2 names the common object. Section 3 says what each paper does to it. Section 4 lists the models of the object and what each one sees and misses. Section 5 tests the deformation-theoretic unification and replaces it by a statement we can prove. Section 6 follows one 2×3 example through every layer. Section 7 separates what is proved from what is computed and what is open.

2 The common object

Let X and Y be finite sets with $|X| = n$ and $|Y| = m$. The *marginal map*

$$M \colon \mathsf{D}(X \times Y) \longrightarrow \mathsf{D}(X) \times \mathsf{D}(Y), \qquad \pi \longmapsto (\pi_X, \pi_Y),$$

sends a joint law to its two margins. Its fibre over (f, g) is the coupling polytope $\mathrm{Cpl}(f, g)$, of dimension $(n-1)(m-1)$ when the margins are positive. Its base, the space of margins, is cut by the *walls* $f(I) = g(J)$, for nonempty proper $I \subset X$ and $J \subset Y$, into *chambers*, on which the combinatorial type of the fibre is constant [5, Prop. 6.6]. It is bounded by the faces of the simplices, where a margin has a zero. With more than two variables the same picture holds for the map from a joint law to its marginals on a cover, and then the fibre can be empty: that is contextuality [9].



Three features of this family carry most of the programme.

(a) *Composition.* A coupling of (f,g) and one of (g,h) glue to a coupling of (f,h). This is the vertical composition of 2-cells [5, §2], it is strictly associative, and the whiskerings with stochastic kernels are not functorial.

(b) *Stratification.* The fibre changes combinatorial type only across walls, and loses dimension only at the boundary. This is what the bands paper calls tearing and breaking [3, Thm. 3.3].

(c) *Positivity.* Laws are nonnegative. Most inequalities in the programme come from here: Fréchet bounds, the emptiness of fibres over cycles, the break margin, the tolerance of a band. Support floors are positivity raised to a policy: a cell floor translates the margins [5, Prop. 6.9].

3 What each paper does

paper	acts on	main structures
Tower [8]	fibres over a fixed outcome base, at three levels	coherent contexts, a strict mixed distributive law, obstructions with certificates, transport along kernels
Conditionals [5]	fibres and their composition	kernels as 1-cells, couplings as 2-cells, the graded shadow, the nerve $\mathsf{D}(Y^{\bullet+1})$, cumulants and the square-zero cone, toric realization
Family [6]	one fibre as a moment polytope	the 2×3 chambers and surfaces, Kähler class = product coupling, disc potentials, stability conditions
Three layers [7]	data seen through operations, probability, outcomes	holonomy, the contextual fraction on cycles, the separation of the three layers
Attribution [2]	polytopes of explanations	attribution sets, precision, mechanism reach, contextual fraction = minimal unmodeled weight
Algebras [1]	the algebra of a fibre	the ring of contingency tables, the gluing algebra $A_f \cong \mathbb{R} \times M_{n-1}(\mathbb{R})$
Bands [3]	the base	bands, the three regimes, entropy on bands, the update monoid, limits, shards, horns and boundaries
Bridges [4]	maps out of the family	entropy and constraint branes, contextuality by default, toric branes, the modular Floer layers
Ambience [8, §10.7], [5, §3, §4]	decisions over the family	support floors as reduced margins, ordered fibres, policies and summaries, support-contextuality, leaf-spring chains, typed obstructions [21, Section 14]

Read top to bottom, the table moves from the fibre over a single base point (Tower), through the structure of fibres and their composition (Conditionals, Family, Algebras), to explanations and data (Attribution, Three layers), to the base and how it moves (Bands), and finally to the maps from the whole family into other geometries (Bridges).

4 Many models of one family

Each language below models the family of section 2. None is complete, and the programme develops several of them side by side. The right-hand column records what the model does not see, or, in the last rows, what has not been tested.

model	what it sees	where	what it misses
linear programming	emptiness of fibres, contextual fraction, attribution sets, certificates	[2, 8]	composition is only a set operation
sheaves and descent	compatible families on covers, global sections, contextuality by default	[9, 14], [4, §4]	quantities: descent is yes or no until the contextual fraction is added
simplicial sets	the nerve $\mathsf{D}(Y^{\bullet+1})$, horn filling	[5, §4], [3, Prop. 6.2]	the Kan condition records failure, not its size
cumulants, A_∞	centred couplings, whiskering defects, the square-zero cone	[5, §5]	cohomology keeps only the marginals; coupling data are homotopical
information geometry	Fisher metric, entropy, maximum-entropy laws, the entropy brane	[4, §§2–3]	the walls: entropy is smooth across them
toric and birational geometry	moment polytopes, Kähler class, walls as variation of GIT	[5, §§6, 10], [6]	positivity of the real points beyond the moment map
Floer theory	nonzero torus branes, leximin couplings, real locus	[4, §§5, 8], [5, §10.4]	composition: gluing is not equivariantly induced [4, Thm. 6.1]
stability conditions	central charges from $\omega_{f,g}$	[6], [5, §§8, 10.3]	hearts and the gluing condition are untested
dynamics	bands, tolerance, tear and break times, the update monoid	[3]	higher structure, which lives on fibres

Floer theory is one row of this table. Its strength is that it singles out couplings: for 2×3 margins, the nonzero torus branes sit where cells are smallest and equal, the leximin coupling carries one in every computed case, and the maximum-entropy coupling carries none when its smallest facet product $f_i g_j$ is unique [4, Thm. 8.2]. Its weakness is composition, which it does not see. The sheaf and simplicial rows see composition and its failure. The information-geometric row sees entropy but not the walls. No row is redundant.

A topos-theoretic model, in which contextuality is the failure of a global element of a presheaf, exists for quantum contextuality [18] and is close to the sheaf row. The programme has not developed it.

5 Deformation theory: what it unifies and what it does not

The proposal is that every paper studies one cotangent complex, with bending, tearing and breaking as its tangent space, its obstruction space and its higher obstructions. Deformation theory has the right shape for a programme about perturbations, so the proposal deserves a test. The test has three parts: what deforms, what its tangent and obstruction spaces are, and whether the regimes match them.

5.1 The natural candidates are rigid

Proposition 5.1 (Rigidity)**.**

(i) The fibre. $\mathrm{Cpl}(f, g)$ *is a polytope in an affine space. At a coupling in its relative interior, the first-order deformations with fixed margins are the matrices with zero row and column sums, a linear space of dimension* $(n-1)(m-1)$*, and there are no obstructions: every first-order*

deformation integrates to a line. At a boundary coupling, the directions that lower a zero cell do not integrate; that is positivity, not a cohomological obstruction.

(ii) The gluing algebra. *$A_f \cong \mathbb{R} \times M_{n-1}(\mathbb{R})$ is separable, so its Hochschild cohomology vanishes above degree 0 [1, Thm. 2.2], [4, Prop. 7.1]. It has no nontrivial deformations and no obstructions.*

(iii) The complex structure. *Each of the five toric surfaces of 2×3 couplings has $H^1(X, T_X) = H^2(X, T_X) = 0$. Its complex structure is rigid and unobstructed.*

Proof. (i) is linear algebra. (ii) is cited. (iii) The surfaces are del Pezzo, so $-K_X$ is ample, and by Serre duality and Akizuki–Nakano vanishing $H^2(T_X) \cong H^0(\Omega^1_X \otimes K_X)^\vee = 0$ [11]. Riemann–Roch for the rank-two bundle T_X on a rational surface gives $\chi(T_X) = 2 + K_X^2 - e(X) = 14 - 2r$ for a toric surface with r rays. By Demazure, $h^0(T_X) = \dim \mathrm{Aut}(X) = 2$ plus the number of Demazure roots [12]. Counting roots gives $6, 4, 4, 2, 0$ for $\mathbb{CP}^2$, $\mathbb{CP}^1 \times \mathbb{CP}^1$, $\mathbb{F}_1$, $\mathrm{Bl}_2\mathbb{CP}^2$ and $\mathrm{Bl}_3\mathbb{CP}^2$, so $h^0 = 8, 6, 6, 4, 2 = \chi(T_X)$ and $h^1(T_X) = 0$. This agrees with Ilten's analysis of smooth toric surfaces [17]. The counts are in `check_architecture.py`. □

So the cotangent complexes of these three objects are trivial in the degrees that matter. (The proposition does not cover the quotients on the walls, which are singular at flip walls, or the boundary fibres; these are where the programme's content sits.) By the proposal's own test, that "if the cotangent complex is trivial, the programme is trivial", the programme would be trivial. It is not: it has walls, chambers, empty fibres and nonzero contextual fractions. So the content of the programme does not live in these deformation spaces. The correspondence "bending $= T^1$, tearing $= T^2$, breaking $= T^{\geq 3}$" fails in particular. Bending is moving the margins within a chamber, which is the motion of the Kähler class described in the next subsection, and tearing and breaking are not obstructions to extending a first-order deformation: they are changes of combinatorial type and of dimension, which happen at finite distance [3, Rem. 3.5].

5.2 What does deform: the symplectic class

The margins do not change the complex structure. They change the symplectic form. By [6, Prop. 2.2], the Kähler class of the toric manifold of $\mathrm{Cpl}(f, g)$ is the product coupling read as divisor coefficients,

$$[\omega_{f,g}] = \sum_{c=(i,j) \text{ facet}} f_i g_j \, [D_c] \in H^2(X; \mathbb{R}).$$

Proposition 5.2 (Margins as Kähler parameters)**.** *For 2×3 couplings the map from margins (f_1, g_1, g_2) to Kähler classes has rank $1, 2, 2, 3, 3$ on the chambers of type $\mathbb{CP}^2$, $\mathbb{CP}^1 \times \mathbb{CP}^1$, $\mathbb{F}_1$, $\mathrm{Bl}_2\mathbb{CP}^2$ and $\mathrm{Bl}_3\mathbb{CP}^2$, where $\dim H^2(X; \mathbb{R}) = 1, 2, 2, 3, 4$. So the margins move the Kähler class over an open subset of the Kähler cone, except for the degree-6 del Pezzo surface: there the six normals sum to zero, the sum of the support numbers is $c_1 \cdot [\omega]$, and on that chamber it equals $f_1 + g_1 + g_2 + (g_3 - f_1) = 1$, so the margins sweep part of the hyperplane $c_1 \cdot [\omega] = 1$ in the four-dimensional Kähler cone. Within a chamber the class moves in the ample cone of one surface; across a wall it passes into the ample cone of a blow-up or blow-down; at the boundary of the space of margins it reaches the boundary of the effective cone.*

Proof. The class is the vector of support numbers modulo linear functions; the ranks are computed in `check_architecture.py`. The chamber structure is that of variation of GIT [5, Prop. 10.5], [20, 13], which for toric quotients is the secondary fan [16, 19]; a single wall crossing for 2×3 is one blow-up or blow-down [6, Prop. 3.3], [4, Prop. 8.1]. □

This is the precise sense in which deformation theory organizes the programme. The deformation parameter is the symplectic class, its moduli space is the Kähler cone of the family of surfaces glued along the walls of the secondary fan, and the three regimes are positions in that space: inside an ample cone (bending), on a wall between two of them (tearing), and at the boundary of the effective cone (breaking).

5.3 The Floer side has its own deformation space

On the Floer side the deformation-theoretic statement is a theorem. Bulk deformations by classes $\mathfrak{b}$ deform the disc potential to $\mathfrak{PO}_{\mathfrak{b}}$, and the Kodaira–Spencer map is a ring isomorphism

$$QH_{\mathfrak{b}}(X;\Lambda_0) \cong \mathrm{Jac}(\mathfrak{PO}_{\mathfrak{b}})$$

[15, Thm. 1.1.1]. So the deformations of the potential are governed by quantum cohomology. For rational 2×3 margins the disc potential is Morse, with as many critical points as the polygon has vertices [4, Thm. 8.2], so $QH(X;\Lambda)$, over the Novikov field, is semisimple, a product of one copy of the field for each nonzero torus brane. For every 3×3 margin the potential is given by the same uncorrected Laurent-polynomial formula, because the semi-Fano corrections vanish [4, Thm. 8.4]; that it is Morse is shown for one example [4, Ex. 8.5]. Identifying this ring with the Hochschild cohomology of the Fukaya category, the object a cotangent-complex proposal would point to, is the subject of [10]. Either way it is a statement about one model, Floer theory, and not about the whole family.

6 One example through every layer

Take $f = (0.4, 0.6)$ and $g = (0.3, 0.43, 0.27)$. Write a coupling by its first-row cells $(\alpha_{11}, \alpha_{12})$; the others follow from the margins. Every number below is recomputed in `check_architecture.py`.

layer	in the example	reference
fibre	$\mathrm{Cpl}(f,g)$ is a pentagon with vertices $(0, \frac{13}{100})$, $(0, \frac{2}{5})$, $(\frac{13}{100}, 0)$, $(\frac{3}{10}, 0)$, $(\frac{3}{10}, \frac{1}{10})$ and area $\frac{1331}{20000}$. Five cells can vanish; $\alpha_{22} \geq f_2 + g_2 - 1 = \frac{3}{100}$ cannot.	[5, §6], [4, Prop. 8.1]
base	The wall values $f_1 - g(J)$ are $\frac{1}{10}, -\frac{3}{100}, \frac{13}{100}, -\frac{33}{100}, -\frac{17}{100}, -\frac{3}{10}$. The tear margin is $\frac{3}{100}$, at the wall $f_1 = g_2$, which is exactly where α_{22} could reach 0; the break margin is $\frac{27}{100}$.	[3, Thm. 3.2]
bands	An elementwise band of half-width δ around (f,g) meets no wall exactly when $\delta < \frac{3}{200}$.	[3, Prop. 3.4]
entropy	The maximum-entropy coupling is the product, $(\frac{3}{25}, \frac{43}{250})$, with entropy 2.5256 bits. The entropy brane meets the conormal brane of the fibre there, transversally.	[4, Thm. 3.1]
algebra	$A_f \cong \mathbb{R} \times \mathbb{R}$, which is semisimple, hence rigid. The ring of contingency tables of the lattice polytope $100 \cdot \mathrm{Cpl}(f,g)$ is its Ehrhart ring, with X as its Proj.	[1, Thms. 2.2, 7.1], [4, Prop. 7.1]

layer	in the example	reference
toric	$X = \mathrm{Bl}_2\mathbb{CP}^2$, with invariant curves of self-intersection $0, -1, 0, -1, -1$ on the cells $11, 12, 13, 21, 23$, and Kähler class $\frac{3}{25}D_{11}+\frac{43}{250}D_{12}+\frac{27}{250}D_{13}+\frac{9}{50}D_{21}+\frac{81}{500}D_{23}$. The function $-\frac{1}{2}H$ is an admissible symplectic potential. $h^0(T_X) = 4$, $h^1 = h^2 = 0$.	[6], [4, Thm. 5.1], proposition 5.1
crossing the wall	Moving g_2 below f_1 makes α_{22} a facet: the pentagon becomes a hexagon, X is blown up to the degree-6 del Pezzo surface, and one more vertex and one more nonzero brane appear.	[6, Prop. 3.3], [4, Prop. 8.1, Thm. 8.2]
Floer	Five nonzero torus branes, all distinct: three over the leximin coupling $(\frac{2}{15}, \frac{2}{15})$, where $\alpha_{11} = \alpha_{12} = \alpha_{13} = \frac{2}{15}$; one over $(\frac{13}{100}, \frac{13}{100})$, where $\alpha_{11} = \alpha_{12} = \alpha_{23}$; one over $(\frac{1}{5}, \frac{1}{10})$, where $\alpha_{12} = \alpha_{13} = \alpha_{21}$. None over the product coupling, whose smallest facet cell $\alpha_{13} = \frac{27}{250}$ is unique.	[4, Thm. 8.2]
real locus	$X^{\mathbb{R}} = \#3\,\mathbb{RP}^2$, Euler characteristic -1, non-orientable; the three (-1)-curves give discs of Maslov index 1, so its Floer theory needs bounding cochains (open).	[4, Prop. 8.6]
nerve	A single table has one context, so it is always compatible. Contextuality needs a cycle of contexts: three pairwise tables on three variables, the boundary of a 2-simplex, the smallest configuration that need not fill.	[5, Prop. 4.5], [3, Prop. 6.2(ii)]
contextuality by default	If the same variable is measured in two contexts with different laws, the fibre condition is replaced by maximal couplings of its copies.	[4, §4]
tower and attribution	Read as a claim about a joint law with these margins, the pentagon is the admissible set at level one; obstructions and certificates come from linear programming; for a horn encoded as a level-one claim the secondary obstruction is at least twice the contextual fraction, with equality when all compatible families are admissible.	[8], [5, Thm. 11.2], [2]
three layers	What this example cannot show: holonomy of operations is invisible to the probability layer.	[7]

The example shows the division of labour. The polytope, its walls and its entropy are probability. The rigidity of X and the motion of its Kähler class are the deformation theory. The three couplings over which branes sit, and the absence of the maximum-entropy coupling among them, are what the Floer model adds. The fact that one table can never be contextual, while three can, is what the sheaf and simplicial models add.

7 Status

statement	status	where
gluing is strictly associative; whiskering by stochastic kernels is not functorial	proved	[5]
2-horns fill, n-horns can fail, the nerve is not Kan	proved	[5, §4], [3, Prop. 6.2]
contextual fraction = minimal unmodeled weight	proved	[2]
coupling polytopes of nondegenerate margins are Delzant; Kähler class = product coupling	proved	[5, Thm. 6.4], [6, Prop. 2.2]
walls are variation-of-GIT walls	proved	[5, Prop. 10.5]
compatibility is lost only through breaking	proved	[3, Thm. 8.1]
entropy brane meets the constraint brane once, transversally, when a positive law satisfies the constraint	proved	[4, Thm. 3.1]
2×3: Morse potential (rational margins), two kinds of brane (off finitely many planes), product coupling excluded (unique smallest facet product)	proved	[4, Thm. 8.2]
3×3: semi-Fano, no disc-potential corrections	proved by computation over all chambers	[4, Thm. 8.4]
natural algebraic and complex deformations are rigid; margins move the Kähler class	proved	propositions 5.1 and 5.2
leximin coupling carries a brane	computed (2970 cases)	[4, Thm. 8.2(iv)]
Floer theory of the real locus beyond monotone cases	open	[4, §8.4]
sheaf model of the entropy brane	open (one isotopy)	[4, Prop. 8.7]
no torus-equivariant correspondence induces gluing by a non-constant stochastic kernel	proved (no-go)	[4, Thm. 6.1]
a topos-theoretic model	not developed	[18]
cell floors are reduced margins: a policy alone can tear or break a system	proved	[5, Prop. 6.9]
floored gluing of the binary triangle: the 3-cycle inequalities with right side $1-8\tau$; support-contextuality	proved	[5, Thm. 4.8]
along a leaf-spring chain certification is path-independent and lost only at summaries	proved	[8, Thm. 10.26]
no order monotone under pooling makes the absolute tower's shift law lax or oplax; coordinate-wise pooling orients the averaging law's single defect at level one	proved	[8, Thm. 13.15]

How to read the programme. Start with section 2. Then read the paper whose model is closest to the reader's own field, using section 4 to find it, and use the example of section 6 as a dictionary between that paper and the others. A statement proved in one model is a statement about the family; whether it transfers to another model is a separate question, and the programme answers it only where the table above says so.

8 Computational checks

All code is at https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/architecture. `check_architecture.py` recomputes the example of section 6 (the polygon, its area, the walls and margins, the band tolerance, the product coupling and its entropy, the self-intersections and Kähler coefficients, the five brane positions with their multiplicities and their distinctness), the Demazure-root counts and Euler characteristics of proposition 5.1, and the ranks of proposition 5.2. It uses the exact routines `newton23.py` and `sweep23.py` of [4, §8], from the folder `fukaya` of the Bridges code, which the script expects next to it as `../bridges/fukaya`.

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